

Wake Forest Summer Internship Networking Reception

Guest Speakers



Julie Freischlag, MD, FACS, FRCSEd(Hon), DFSVS, MAMSE

Chief Academic Officer & CEO, Atrium Health WFB

Executive VP of Advocate Health, Executive VP for Health Affairs at WFU

Email: jfreisch@wakehealth.edu

Dr. Julie A. Freischlag is a distinguished leader in the medical field, currently serving as the chief executive officer and chief academic officer of Atrium Health Wake Forest Baptist, chief academic officer and executive vice president of Advocate Health, and executive vice president for health affairs at Wake Forest University. In her roles, she oversees research, discovery, and educational programs across multiple institutions within the Advocate Health

system and manages an annual operating budget exceeding \$4 billion.

Dr. Freischlag's extensive experience includes serving as dean of Wake Forest University School of Medicine and vice chancellor for Human Health Sciences at UC Davis. With over three decades of leadership in patient care and surgical departments at prominent hospitals, she has also made significant contributions to medical education and training. A specialist in thoracic outlet syndrome,

Dr. Freischlag has been recognized with numerous accolades, including the 2023 Lifetime Achievement Award from the Society for Vascular Surgery and election to the National Academy of Medicine. Her leadership roles extend nationally, having served as the 2021-2022 President of the American College of Surgeons and holding positions on various boards, such as the Association for American Medical Colleges and the National Institute of Health Clinical Center Research Hospital Board. In November 2024, she will assume the role of chair-elect of the Association of American Medical Colleges' board of directors.



James Monaco, MD

Scientist in Statistical Arbitrage

Budo Group

Email: jamespmonaco@gmail.com

James Monaco received a BS in computer engineering and an MS in electrical engineering from Texas A&M University in 1996 and 1998. He received his PhD in electrical engineering from the University of Texas at Austin in 2007.

Dr. Monaco began his career designing infrared tracking systems for military aircraft at Texas Instruments. He later became the first employee of VuCOMP, a medical imaging startup that created devices for detecting cancer in mammograms. Returning to academia, Dr. Monaco spent five years as a research professor at Rutgers University. Here he studied the automated prognosis and therapeutic prediction of disease using pathology images. This research culminated in the co-founding of the company IbRiS. In 2014 Inspirata Inc. acquired IbRiS, and Dr. Monaco joined as the head of research and development.

Six years ago, Dr. Monaco changed careers and now develops methods for modeling futures markets in the financial industry.

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Program Directors & Event Mentors



Ashley Weaver, PhD

Summer Research Internship Program Co-Director (BME, CALIBIR, & eREU KEEN)
Recruitment Director, WFU-VT SBES Graduate Program
Associate Professor, Biomedical Engineering
Email: asweaver@wakehealth.edu

Research Focus: Accidents - Traffic, Wounds and Injuries, Tomography, X-Ray, Computed Lumbar Vertebrae, Aging Fractures - Bone



Ryan McGinnis, PhD

Summer Research Internship Program Co-Director (BME & eREU KEEN)
Director, Center for Remote Patient Monitoring
Associate Professor, Biomedical Engineering
Email: rmcginni@wakehealth.edu

Research Focus: Mental Health, Biomedical Engineering, Machine Learning, Biomechanics, Rehabilitation, Biomedical Data Science, User-Centered Design, Wearables



Metin Gurcan, PhD

Summer Research Internship Program Co-Director (CALIBIR)
Director, Center for Artificial Intelligence Research (CAIR)
Professor, Internal Medicine, Pathology, BME
Email: metin.gurcan@advocatehealth.org

Research Focus: Image Interpretation Computer-Assisted, Image Processing Computer-Assisted, Pattern Recognition - Automated, Algorithms, Biomedical Informatics, Artificial Intelligence



Arezoo Movaghar, PhD

Summer Research Internship Program Co-Director (CALIBIR)
Assistant Professor, Pediatrics
Center for Artificial Intelligence Research (CAIR)
Email: arezoo.movaghar@advocatehealth.org

Research Focus: Artificial Intelligence, Biomedical Informatics, Electronic Health Records, Social Determinant of Health, Diagnostic and Health Disparities, Neurological Disorders, Rare Diseases

Wake Forest Summer Internship Networking Reception

Program Event Mentors

**Nicholette Allred, PhD**

Professor, Biochemistry

Program Mentor: Center for Artificial Intelligence Research / CALIBIR

Email: nallred@wakehealth.edu

Research Focus: Genetics, Type 2 Diabetes, Insulin Resistance, Metabolomics

**Kristen Beavers, PhD**

Professor, Gerontology and Geriatrics, Internal Medicine

Program Mentor: Biomedical Engineering (BME)

Email: kristen.beavers@advocatehealth.org

Research Focus: Aging, Diabetes/Obesity/Metabolism, Biomedical Imaging, Musculoskeletal Care, Exercise & Diet Interventions, Body Composition

**Karan Devane, PhD**

Assistant Professor, Biomedical Engineering

Program Mentor: Biomedical Engineering (BME) & Draelos Summer Scholars

Email: karan.devane@advocatehealth.org

Research Focus: Human Body Modeling, Injury Biomechanics, Spaceflight Safety, Crash Reconstruction, Accident Data Analysis, Vehicle and Infrastructure Safety

**Scott Gayzik, PhD**

Professor, Biomedical Engineering

Director, WFU-VT SBES Graduate Program

Email: francis.gayzik@advocatehealth.org

Research Focus: computational injury biomechanics & imaging applications to create human body computational models for industry, government, & academic applications.

**Omaditya (Goldey) Khanna, MD**

Assistant Professor, Neurosurgery

Program Mentor: Center for Artificial Intelligence Research / CALIBIR

Email: okhanna@wakehealth.edu

Research Focus: Brain & Spinal Tumor, Cerebrovascular Neurosurgery, Interventional Neuroradiology, Skull Base

**Da Ma, PhD**

Assistant Professor, Gerontology and Geriatric Medicine

Program Mentor: Center for Artificial Intelligence Research / CALIBIR

Email: da.ma@advocatehealth.org

Research Focus: Medical Image Analysis, Neuroimaging, Alzheimer's Disease and Related Dementias, AI Applications in Medical Imaging, Neuroimaging Genomics

**Mohammad Moghimi, PhD**

Assistant Professor, Biomedical Engineering

Program Mentor: CALIBIR & Draelos Summer Scholars

Email: seyyed.moghimi@advocatehealth.org

Research Focus: Early Detection of Cancer, Translational Research, Biosensors, Wearable Devices, Microelectromechanical Systems, Flexible Bioelectronics

**Mostafa Rezapour, PhD**

Assistant Professor, Wake Forest Institute for regenerative Medicine

Program Mentor: Center for Artificial Intelligence Research / CALIBIR

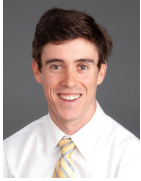
Email: mrezapou@wakehealth.edu

Research Focus: Artificial Intelligence, Biomedical Informatics, Machine Learning, Medical Image Analysis, Bioinformatics, Statistics

**Jillian Urban, PhD, MPH**

Assistant Professor, Biomedical Engineering
 Program Mentor: Biomedical Engineering (BME)
 Email: jurban@wakehealth.edu

Research Focus: Public Health, Epidemiology, Traumatic Brain Injury, Sports Injury, Biomechanics, Head Impact Exposure

**Zach Hostetler, PhD**

Research Engineer, Elemance LLC.
 Program Mentor: eREU KEEN – Elemance LLC
 Email: zach.hostetler@elemance.com

Elemance, LLC, an engineering software and services company, explores the interaction of technologies with the human body. Elemance enables nuanced design through finite element analysis and other data science-based approaches.

**Emilie Duchesneau, PhD, MSPH**

Assistant Professor, Epidemiology and Prevention
 Program Mentor: Cardiovascular & Aging Internship
 Email: educhesn@wakehealth.edu

Research Focus: Epidemiologic methods, cancer, aging, healthcare databases

**Justin Moore, PhD, MS**

Professor, Implementation Science, Public Health Sciences
 Wake Foerst University School of Medicine
 Email: jusmoore@wakehealth.edu

Research Focus: Exercise, Motor Activity, Obesity, Health Promotion, Health Behavior

**William T. Booth, II MLS (ASCP), PhD**

Assistant Professor, Clinical Chemistry
 Winston-Salem State University
 Email: boothwt@wssu.edu

Research Focus: Elucidation of the biochemical details of biological systems

**Patricia Smyre, PhD**

Program Manager III
 Maya Angelou Research Center for Healthy Communities (MARCH)
 Wake Forest University School of Medicine
 Email: psmyre@wakehealth.edu

The **Center for Personal and Professional Development (CPPD)** supports Wake Forest University School of Medicine students' personal and professional development while also equipping them with skills that promote personal reflection, lifelong learning, and collaboration.

**Cheraton Love, PhD**

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**Keyma Clark, PhD**

Coach, Center for Personal and Professional Development (CPPD)
 Email: ktclark@wakehealth.edu

**Kendall Freeman, MS, MFT**

Coach, Center for Personal and Professional Development (CPPD)
 Email: kefreema@wakehealth.edu

**John James, M.Div**

Coach, Center for Personal and Professional Development (CPPD)
 Email: jcjamess@wakehealth.edu

Additional Program Mentors (Unable to attend)



Oguz Akbilgic, PhD

Associate Professor, Cardiovascular Medicine

Program Mentor: Center for Artificial Intelligence Research / CALIBIR

Email: oakbilgic@wakehealth.edu

Research Focus: Artificial Intelligence, Biomedical Informatics, Clinical Informatics, Machine Learning, Statistical Modeling



Philip Brown, PhD

Associate Professor, Biomedical Engineering

Program Mentor: Biomedical Engineering (BME)

Email: phibrown@wakehealth.edu

Research Focus: anatomical modeling, mechanical design, biomechanical experimentation, biomechanical robotics, surgical robotics, FEA, and additive manufacturing of medical models, devices, & implants



Ellen McGinnis, PhD

Assistant Professor, Social Sciences and Health Policy

Program Mentor: Biomedical Engineering (BME), eREU KEEN – BioBe

Email: ewmcginn@wakehealth.edu

Research Focus: Public Health Sciences & Pediatrics



Jared Weis, PhD

Assistant Professor, Biomedical Engineering

Program Mentor: Biomedical Engineering (BME)

Email: jweis@wakehealth.edu

Research Focus: integration of computational modeling and non-invasive imaging methodologies to explore the biophysics of cancer



Ibrahim Karabayir, PhD

Assistant Professor, Internal Medicine & Cardiovascular Medicine

Program Mentor: Center for Artificial Intelligence Research / CALIBIR

Email: ikarabay@wakehealth.edu

Research Focus: Artificial Intelligence, Biomedical Informatics, Clinical Informatics, Machine Learning, Data Science, Clinical Decision Making



Nathaniel O'Connell, PhD

Assistant Professor, Biostatistics and Data Science

Program Mentor: Center for Artificial Intelligence Research / CALIBIR

Email: n.oconnell@wakehealth.edu

Research Focus: cancer phase I trial methodology and machine learning prediction modeling



Ellen Quillen, PhD

Assistant Professor, Molecular Medicine

Program Mentor: Center for Artificial Intelligence Research / CALIBIR

Email: equillen@wakehealth.edu

Research Focus: Skin Pigmentation, Neoplasm Metastasis, Musculoskeletal Diseases, Proteomics, Genomics, Radiation Injury, Diabetes/Metabolism, Epigenetics, Admixture Mapping



Jaime Lynn Speiser, PhD, MSc

Assistant Professor, Biostatistics & Data Science

Program Mentor: Center for Artificial Intelligence Research / CALIBIR

Email: jspeiser@wakehealth.edu

Research Focus: Public Health Sciences, Liver Failure, Models Theoretical, Models Statistical, Aging, Predictive Analytics



Atalie Thompson, MD, MPH

Assistant Professor, Ophthalmology

Program Mentor: Center for Artificial Intelligence Research / CALIBIR

Email: atathomp@wakehealth.edu

Research Focus: Healthcare Disparities, Alzheimer's Disease and Related Dementias, Public Health Innovations, Artificial Intelligence Applications in Medical Imaging, Age-Related Eye Diseases, Factors Impacting Patient Mobility, Ophthalmology

**Adam Hall, PhD**

Professor, Biomedical Engineering

Program Mentor: Biomedical Engineering (BME)

Email: arhall@wakehealth.edu

Research Focus: DNA, Nanotechnology, Cancer, Biomarkers, Glycans

**Kristen Nicholson, PhD**

Assistant Professor, Orthopaedic Surgery & Rehabilitation

Program Mentor: Biomedical Engineering (BME)

Email: kfnichol@wakehealth.edu

Research Focus: Upper Extremity Biomechanics, Motion Analysis, Pitching Biomechanics

**Joel Stitzel, PhD**

Professor, Biomedical Engineering

Program Mentor: Biomedical Engineering (BME)

Email: jstitzel@wakehealth.edu

Research Focus: Models - Biological, Wounds And Injuries, Finite Element Analysis, Thoracic Injuries, Brain Concussion, Concussion in Sports, Aerospace Medicine, Motor Vehicle Crashes, Computational Methods

**Jeff Willey, PhD**

Professor, Radiation Oncology

Program Mentor: Biomedical Engineering (BME)

Email: jwilley@wakehealth.edu

Research Focus: Whole-Body Irradiation, Knee Joint, Space Flight, Hindlimb Suspension, Osteoporosis

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REU & CALIBIR Alumni - Event Mentors



Zach Hostetler, PhD

Research Engineer, Elemance LLC.
Program Mentor: eREU KEEN – Elemance LLC
Email: zach.hostetler@elemance.com
REU Class of 2017



Grace Krell, BS, MBA

Senior Certification Engineer, Collins Aerospace
Founder & CEO of SweetSip
Email: grace.grk@gmail.com
REU Class of 2021



Hannah Weinbaum, BS

Computational Analyst at Corvid Technologies, Missile Defense Systems (MDS)

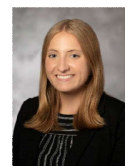
Email: hannah.weinbaum@corvidtec.com
REU Class of 2022



Elsa King, BS

PhD Student, Virginia Tech - Wake Forest University SBES

Email: elsa.king@wfusm.edu
REU Class of 2022 & WFIRM Summer Scholar of 2023



Molly Ehrig, MB

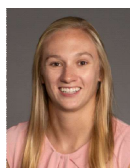
Biostatistician II, Wake Forest University School of Medicine

Email: mehrig@wakehealth.edu
REU & CALIBIR Class of 2023



Samantha Schnell, BS

Computer Science Research Assistant, University of North Carolina at Greensboro (UNCG)
Data Science and Biostatistics Intern, Wake Forest School of Medicine
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REU & CALIBIR Class of 2023



Lauren Washco, BS

PhD Student - Molecular Medicine and Translational Science, Wake Forest University School of Medicine

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REU Class of 2023

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Program Coordinators & Administrators



Claudia Olivier, PhD

Program Director, Training in Research Affecting Child-Health (TRAC)

Hypertension & Vascular Research Center

Email: colivier@wakehealth.edu



Joan Schanck, MPA

Chief Education Program Officer, Wake Forest Institute for Regenerative Medicine (WFIRM)

WFIRM Summer Scholars Program

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Lydia Durr

Research Administrative Coordinator II, Hypertension & Vascular Research Center

Training in Research Affecting Child-Health (TRAC), Cardiovascular & Aging Internship

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Thea Marie Smith

Program Coordinator I, Biomedical Engineering

Summer Research Internship Programs: BME, eREU KEEN, & Draelos Summer Scholars

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Latrice Harris

Associate Project Manager, Center for Artificial Intelligence Research (CAIR)

Program: Culturally Augmented Learning in Biomedical Informatics Research (CALIBIR)

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Jaelyn Holmes

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Lisa Harris

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Ruth Morales

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Kelly McNeal

Associate Administrative Coordinator, Cancer Biology

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